adapted on
Lynford L. Goddard
Brian T. Cunningham

# ECE 329

# Fields and Waves I

Prof. Victoria Shao
Teaching Associate Professor



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# Prof. Victoria Shao

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**Research interests**

- Electromagnetic compatibility/interference (EMC/EMI)
  - ECE 498 Engineering EMC – Prof. Victoria Shao
- Signal and power integrity analysis (SI/PI)
  - ECE 546 Advanced Signal Integrity – Prof. Jose Schutt-Aine
- Computational electromagnetics (CEM)
  - ECE 540 Computational Electromagnetics – Prof. Jianming Jin
- High-power microwave (HPM)

# Outside Classroom

- Two kids (12th grade and 3rd grade)
- Playing Volleyball
- Watch Sci Fi movies and criminal TV series
- Have a dog named Molly





# Course Organization

- Section X: MWF 12:00pm-12:50pm
- Course website
  - https://courses.engr.illinois.edu/ece329/index.html
  - Syllabus
  - Homework due **Thursday** at **11:59pm**
  - Grading policy
  - Canvas, discussion forum, Homework Solutions, Slides
  - Homework assignment on Gradescope
- 3 mid-terms and 1 final
- Office hour: Thursday 11am-12pm
- Prerequisites: MATH 241, PHYS 212, ECE 210
- Follow up course: ECE 350

# Course Organization

- Grade breakdown

| | |
|---|---|
| Homework | 15% of total |
| Midterm Exams (x3) | 60% of total |
| Final Exam | 25% of total |

- Midterm time
  - Exam 1 Sep. 21 Mon 7:00-8:15pm
  - Exam 2 Oct. 19 Mon 7:00-8:15pm
  - Exam 3 Nov. 16 Mon 7:00-8:15pm

# How to get an A in ECE329

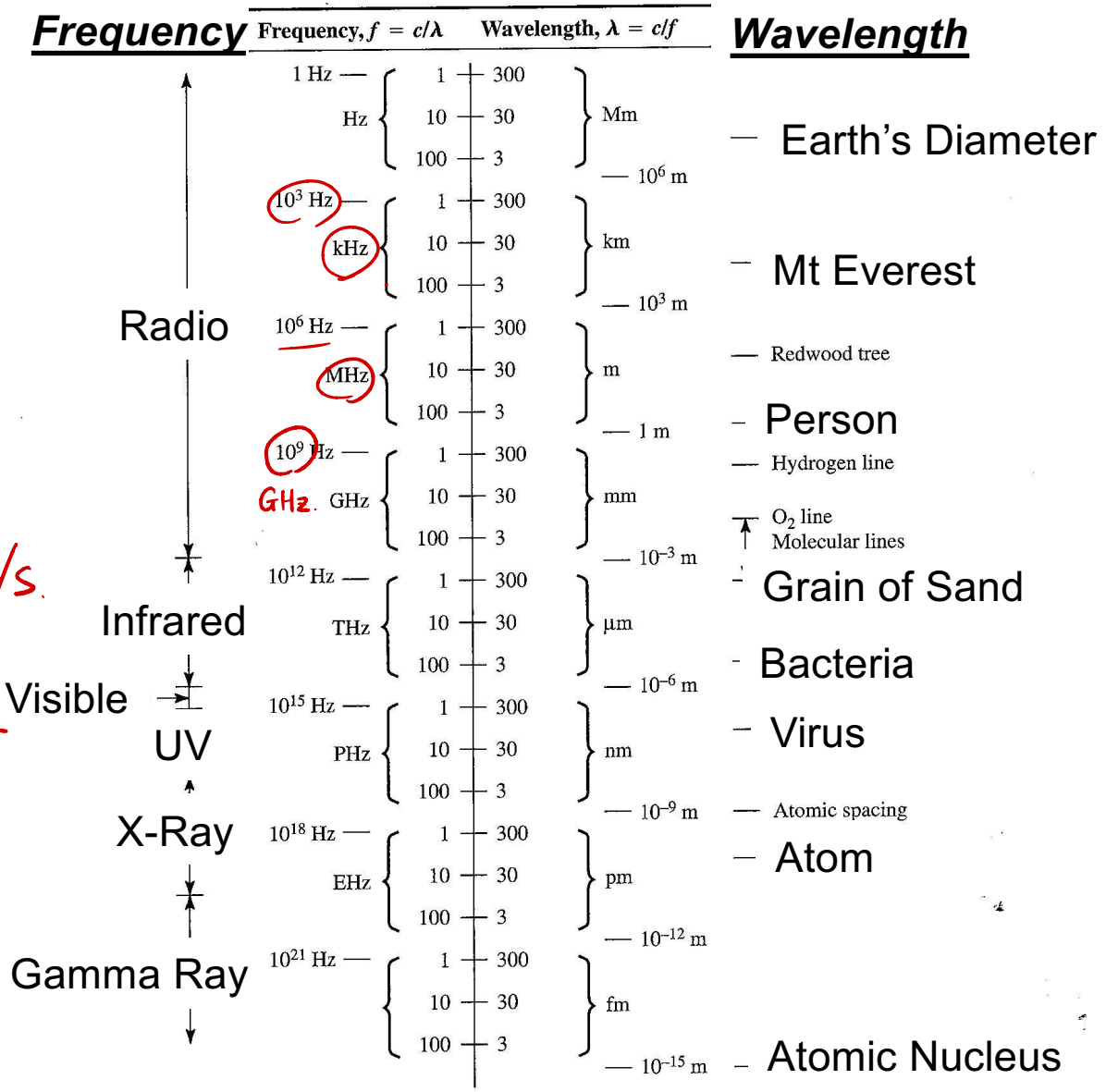
- ***Time Management.*** Allocate 10 hrs/wk of *regularly scheduled times* in the week outside of class for 329:
  - 30min for **reading** of textbook **before** each class
  - 30min for **studying** online notes **before** each class
  - 30min for **studying** my notes **between** classes
  - 5-6 hrs/wk for HWs
    - In a semester, all lectures total only 32.5hrs, which is less than 1 week at a job! It's up to you to put in the time to learn
- Get a 1" binder (organize lecture notes/HWs/exams)
- Start assignments early. **Do all problems by yourself first.** If you get stuck, form study groups to work on problems together but **ALWAYS** write-up and submit **YOUR OWN** solutions. Do not blindly copy.
- Ask questions and come to office hours if you get stuck. Don't let confusion snowball

# How to get an A in ECE329

- ***Practice doing problems.*** Get comfortable with the math manipulations and associated physical meaning, and you will find exam problems to be easier
  - HW problems
  - Example problems worked in lecture
  - Example problems worked in online class notes
  - Old exam problems
  - Office hours
- ***Review your prerequisites.***
  - Vector calculus, line/surface integrals: Math 241
  - Linear circuit, system analysis, phasors: ECE 110/210
  - Electric and magnetic fields: Physics 212
- ***Come to class!!***
  - I will discuss topics to be emphasized on exams and give hints about how to approach the more difficult homework problems.

# EM Spectrum

**TABLE 1-1**
**Frequencies and wavelengths of the electromagnetic spectrum from almost dc to gamma-rays**




$\lambda f = c$

$3 \times 10^8$ m/s

300 m/µs

# Speed of Light

| Distance Travelled | Travel Time |
|---|---|
| Across a virus | $100 \times 10^{-18}$ sec |
| 300 meters | 1 microsecond |
| NY to LA | 13 milliseconds |
| Around earth | 0.133 sec |
| Earth to Moon | 1.2 sec |
| Earth to Sun | 8.3 minutes |
| Earth to Mars | 3-21 minutes |
| Sun to nearest star | 4 years |
| Diameter of our galaxy | 100,000 years |
| Edge of known universe | 15 billion years |

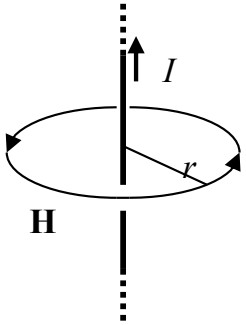
c ≡ 299,792,458 m/s (used to define the meter!)

# Brief History of EM

Charles Coulomb in 1785 demonstrated how electric charges repel one another



Andre Marie Ampere discovered that an electric current produces a magnetic field in 1820





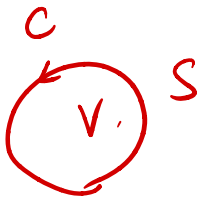

Michael Faraday in 1831 showed that since an electric current could produce a magnetic field, a changing magnetic field can produce an electric current. "Principle of Induction" used for first electric generators.



# James Clerk Maxwell (1831-1879)

- 1855-1868 - formulates field equations for electromagnetism. Predicts existence of EM wave propagation and the speed of light. Shows theoretical possibility of generating electromagnetic radiation
- 1873: publishes *Treatise on Electricity and Magnetism*






# Maxwell's Equations

## Integral form

## Differential Form

| | Integral form | | Differential Form |
|---|---|---|---|
| Faraday | $\oint_C \mathbf{E}\cdot d\mathbf{L} = -\frac{d}{dt}\int_S \mathbf{B}\cdot d\mathbf{S}$ | | $\nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}$ |
| Ampere | $\oint_C \mathbf{H}\cdot d\mathbf{L} = \int_S \mathbf{J}\cdot d\mathbf{S} + \frac{d}{dt}\int_S \mathbf{D}\cdot d\mathbf{S}$ | | $\nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t}$ |
| Gauss | $\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_{Vol} \rho\, dv$ | $\rho = \frac{Q}{V}$ | $\nabla\cdot\mathbf{D} = \rho$ |
| Gauss | $\oint_S \mathbf{B}\cdot d\mathbf{S} = 0$ | | $\nabla\cdot\mathbf{B} = 0$ |

$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$

| | |
|---|---|
| **E** | Electric Field |
| **H** | Magnetic Field |
| **D**=ε**E** | "Displacement Flux Density" |
| **B**=μ**H** | "Magnetic Flux Density" |
| **J** | Current Density |
| ρ | Charge Density |

$I/A = J$

# Units and Constants

Electric field **E**: V/m

Magnetic field **H**: A/m

Displacement flux density **D**: $\varepsilon E$ $C/m^2$ $\Rightarrow \vec{H}, \vec{B}$

Magnetic flux density **B**: T, $wb/m^2$

Current density ***J***: wire A surface A/m volume $A/m^2$

Charge density **ρ**: dot C line of charge C/m $\Rightarrow \vec{E}, \vec{D}$

surface $C/m^2$

volumetric $C/m^3$

$$\varepsilon_o \approx 8.854\times10^{-12}\ F/m$$

$$\mu_o = 4\pi\times10^{-7}\ H/m$$

$$c \approx 3\times10^{8}\ m/s$$

$\downarrow \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

# Sections 1.1-1.2

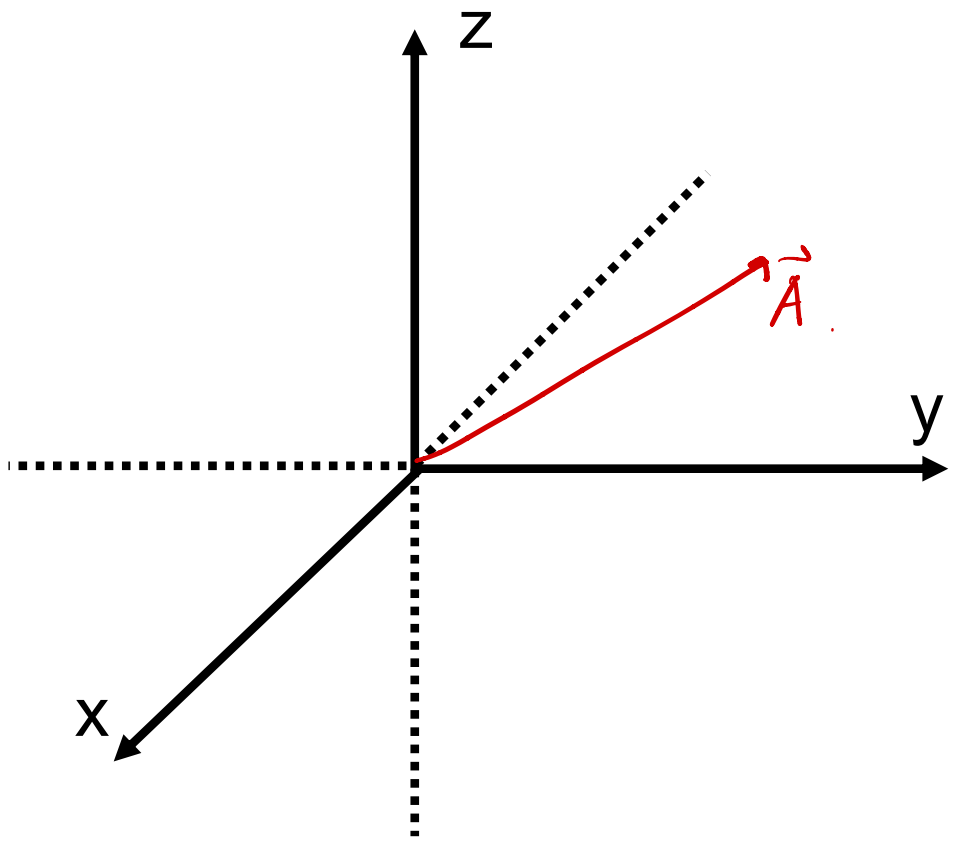
## Vector Algebra
## Cartesian Coordinates
## Differential Length Vector
## Differential Surface Vector

Adapted from Prof. Cunningham's Notes

# Today's Topics

- Review
  - Scalars (numbers) and vectors
  - Unit vectors
  - Vector addition & subtraction
  - Magnitude
  - Dot product
  - Cross product
- New Topics
  - Differential length vector $d\vec{\ell}$
  - Differential surface vector $d\vec{s}$

# Cartesian Coordinate System




Unit vectors

$\hat{a}_x \quad \hat{a}_y \quad \hat{a}_z$

$\hat{x} \quad \hat{y} \quad \hat{z}$

Notation

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z$$

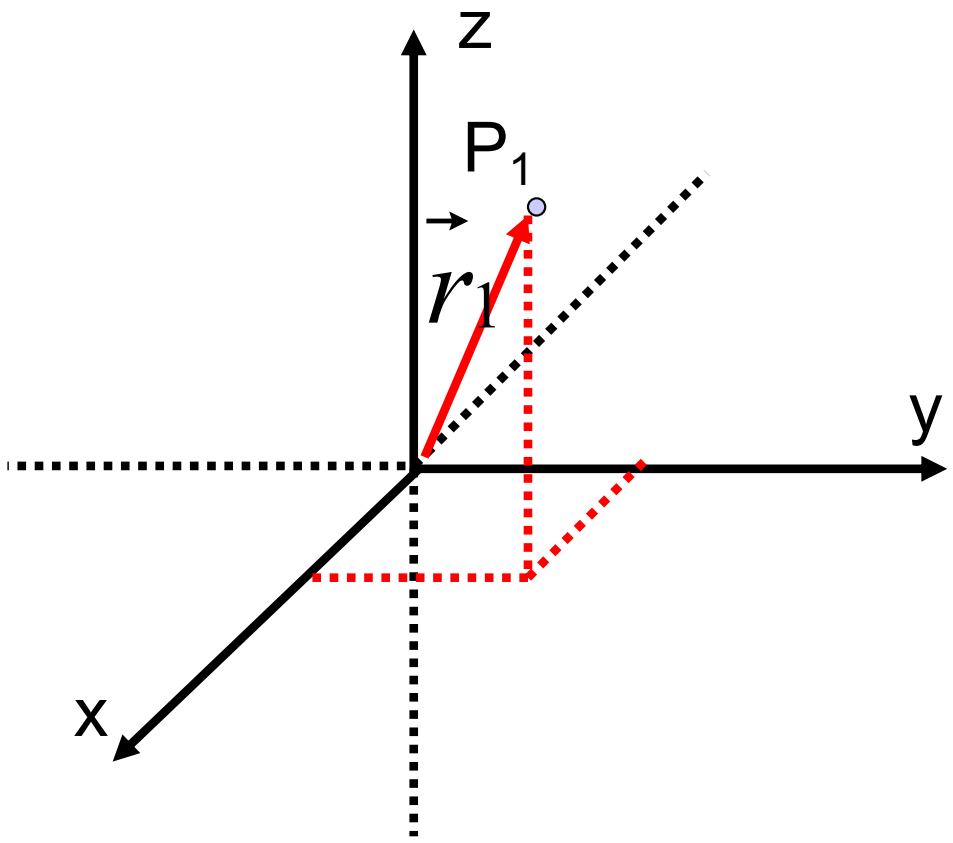
Alternate Notation

$$\vec{A} = < A_x, A_y, A_z >$$

$$\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$$

(Review at home)

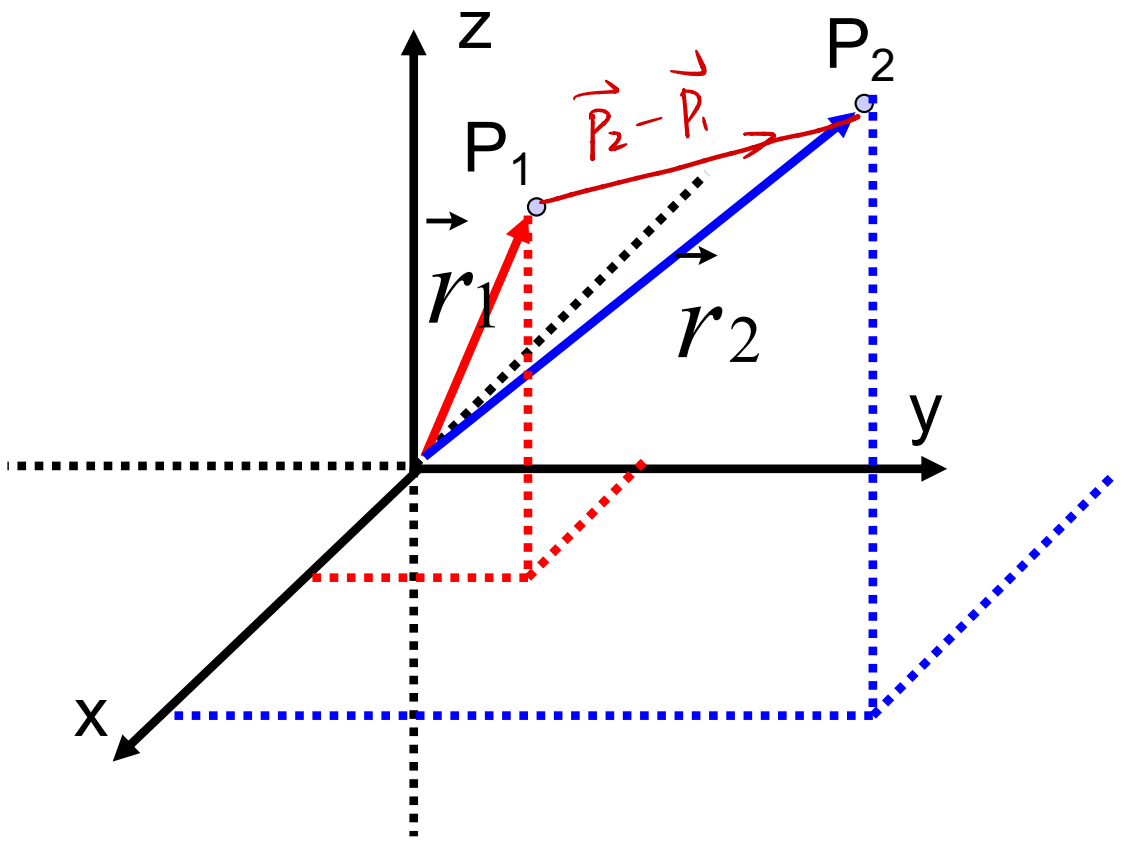
# Vector From Origin to a Point




$P_1$: $(x_1, y_1, z_1)$

(Review at home)

# Vector Between Two Points

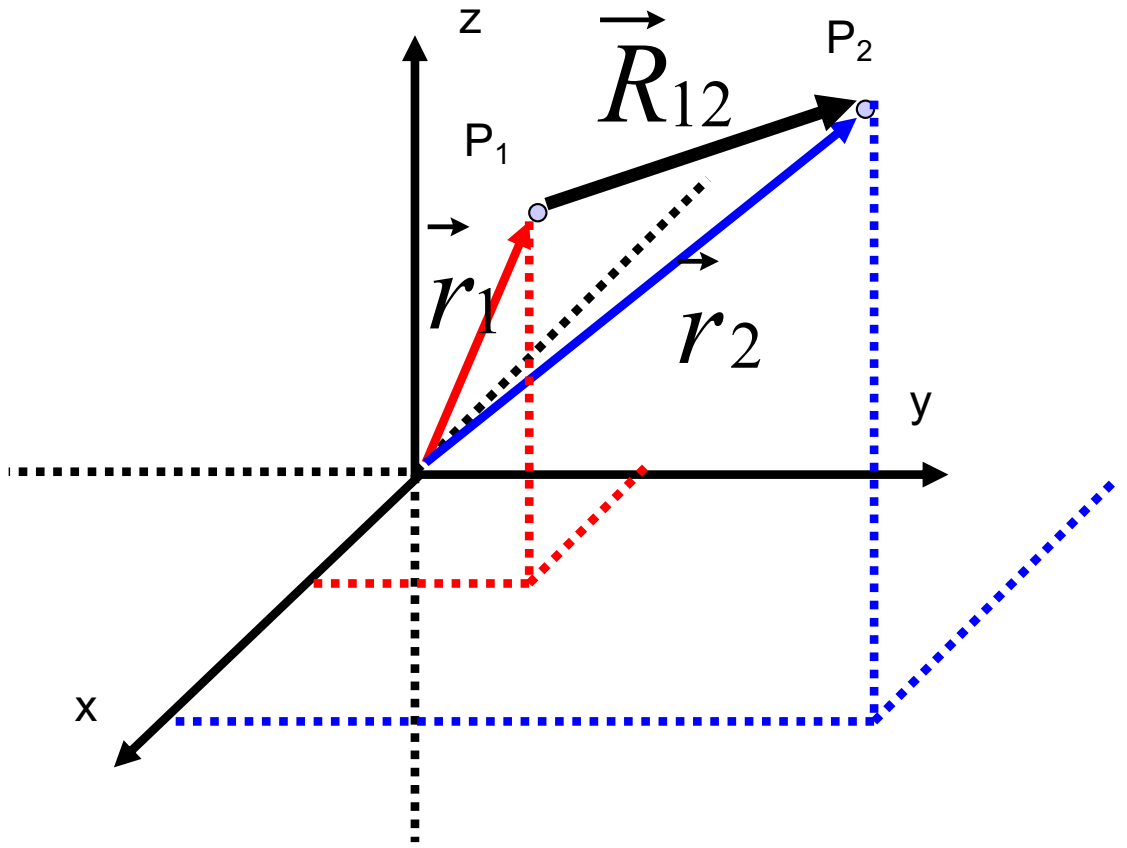



$P_1$: $(x_1, y_1, z_1)$

$P_2$: $(x_2, y_2, z_2)$

(Review at home)

# Vector Between Two Points




$$\vec{R}_{12} = (FinalPosition) - (InitialPosition)$$

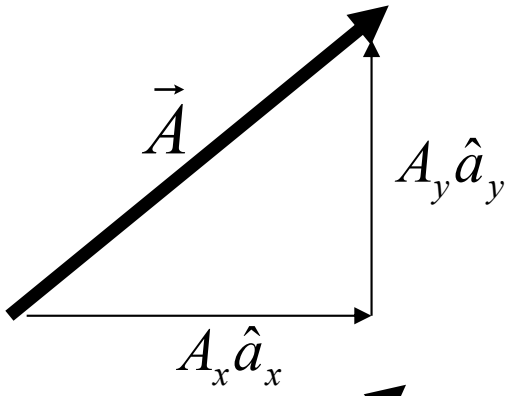
$$\vec{R}_{12} = (x_2 - x_1)\hat{a}_x + (y_2 - y_1)\hat{a}_y + (z_2 - z_1)\hat{a}_z$$

# (Review at home)
# Vector Math

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

## Magnitude of **A**

$$\left|\vec{A}\right| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

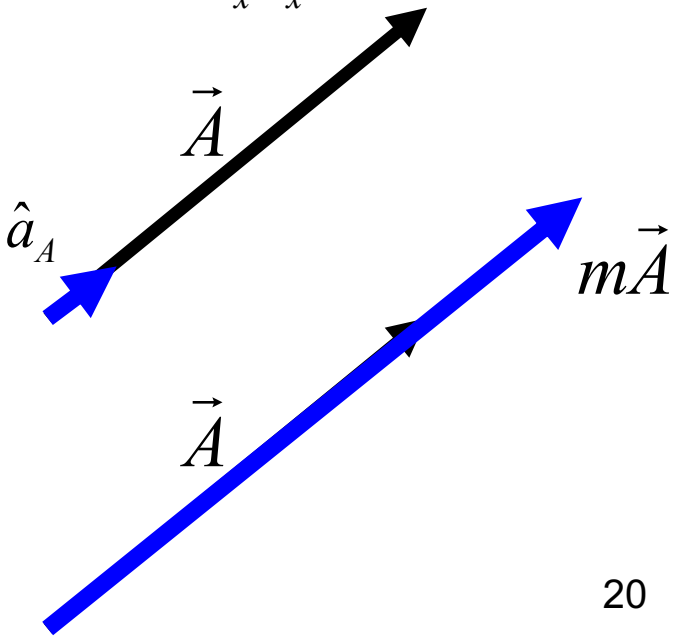



## Unit vector in direction of **A**

$$\hat{a}_A = \frac{\vec{A}}{\left|\vec{A}\right|} = \frac{A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

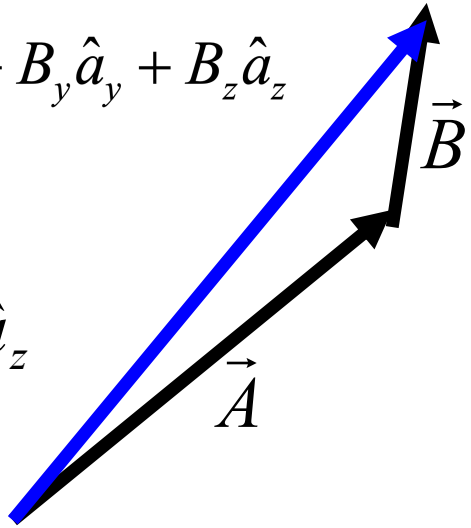
## Multiplication of **A** by a scalar

$$m\vec{A} = mA_x\hat{a}_x + mA_y\hat{a}_y + mA_z\hat{a}_z$$



# (Review at home)
# Vector Math

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

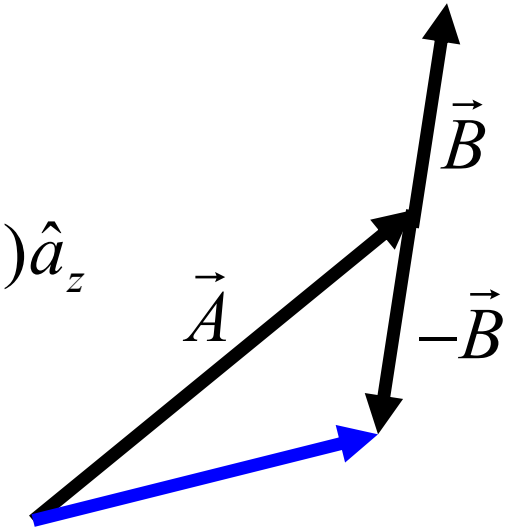



## Addition of **A** and **B**

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{a}_x + (A_y + B_y)\hat{a}_y + (A_z + B_z)\hat{a}_z$$

## Subtraction of **A** and **B**

$$\vec{A} - \vec{B} = (A_x - B_x)\hat{a}_x + (A_y - B_y)\hat{a}_y + (A_z - B_z)\hat{a}_z$$

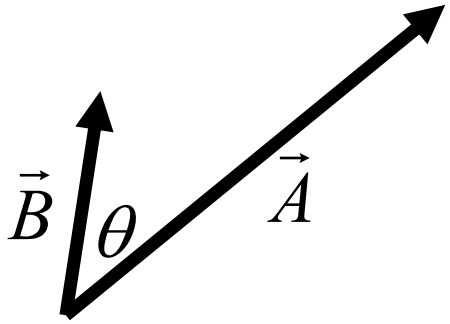


# (Review at home)
# Dot Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

## Dot Product of **A** and **B**

$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\cos\theta$$



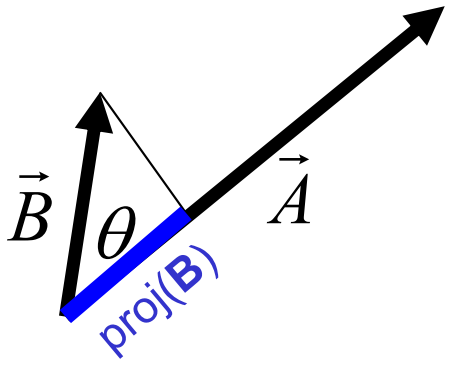

Resulting quantity is a SCALAR

## Physical Meaning

(Magnitude of **A**)*(Projection of **B** onto **A**)

or (Magnitude of **B**)*(Projection of **A** onto **B**)

$$\vec{A} \bullet \vec{B} = \vec{B} \bullet \vec{A}$$



(Review at home)

# Dot Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

## Dot Product of **A** and **B**

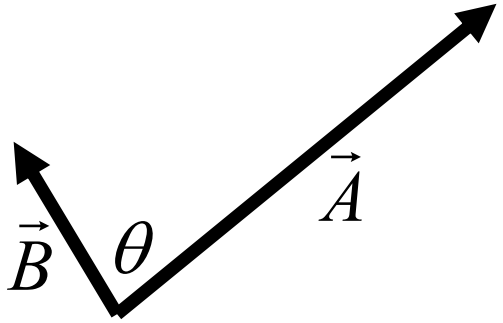
$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\cos\theta$$

$$\vec{A} \perp \vec{B}$$

$$\cos\theta = 0$$

$$\vec{A} \bullet \vec{B} = 0$$

$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = 0$$

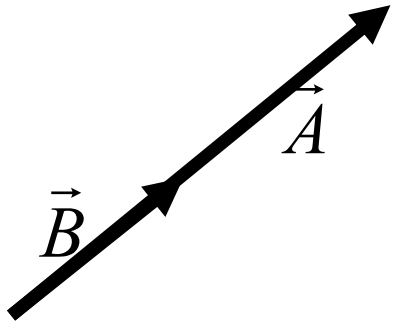



$$\vec{A} \| \vec{B}$$

$$\cos\theta = 1$$

$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$

$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$



(Review at home)

# Dot Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = 0$$

$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$

## Dot Product of **Unit Vectors**

$$\hat{a}_x \bullet \hat{a}_x = 1 \quad \hat{a}_y \bullet \hat{a}_x = 0 \quad \hat{a}_z \bullet \hat{a}_x = 0$$
$$\hat{a}_x \bullet \hat{a}_y = 0 \quad \hat{a}_y \bullet \hat{a}_y = 1 \quad \hat{a}_z \bullet \hat{a}_y = 0$$
$$\hat{a}_x \bullet \hat{a}_z = 0 \quad \hat{a}_y \bullet \hat{a}_z = 0 \quad \hat{a}_z \bullet \hat{a}_z = 1$$

$$\vec{A} \bullet \vec{B} = \left(A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z\right) \bullet \left(B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z\right)$$

$$\vec{A} \bullet \vec{B} = A_xB_x + A_yB_y + A_zB_z$$

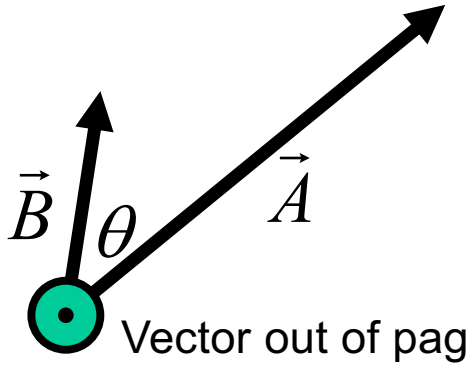
Scalar

# (Review at home)
# Cross Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

## Cross Product of **A** and **B**

Resulting quantity is a VECTOR




Vector out of page pointing at the classroom

### Magnitude

$$\vec{A} \times \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\sin\theta$$

### Direction

Perpendicular to BOTH **A** and **B**

Two possible vectors satisfy this condition

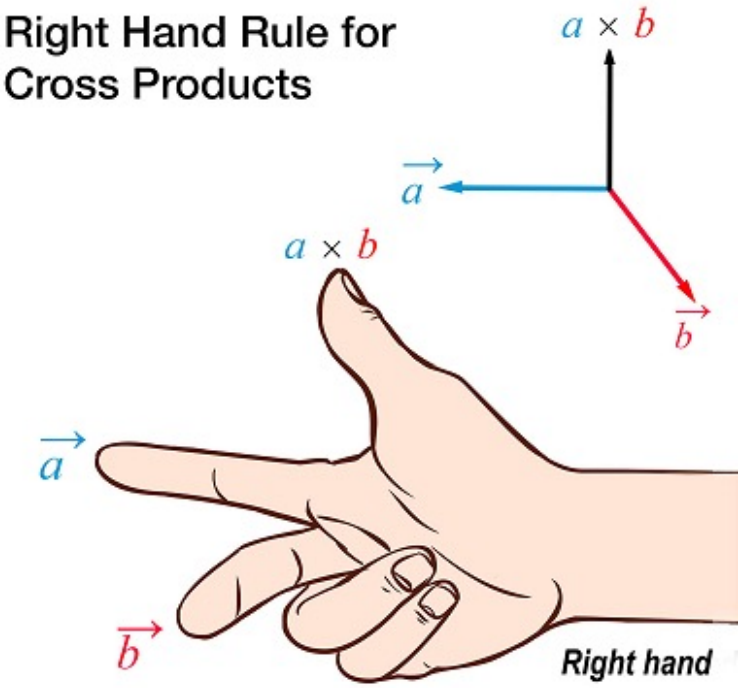
Determined using the RIGHT HAND RULE

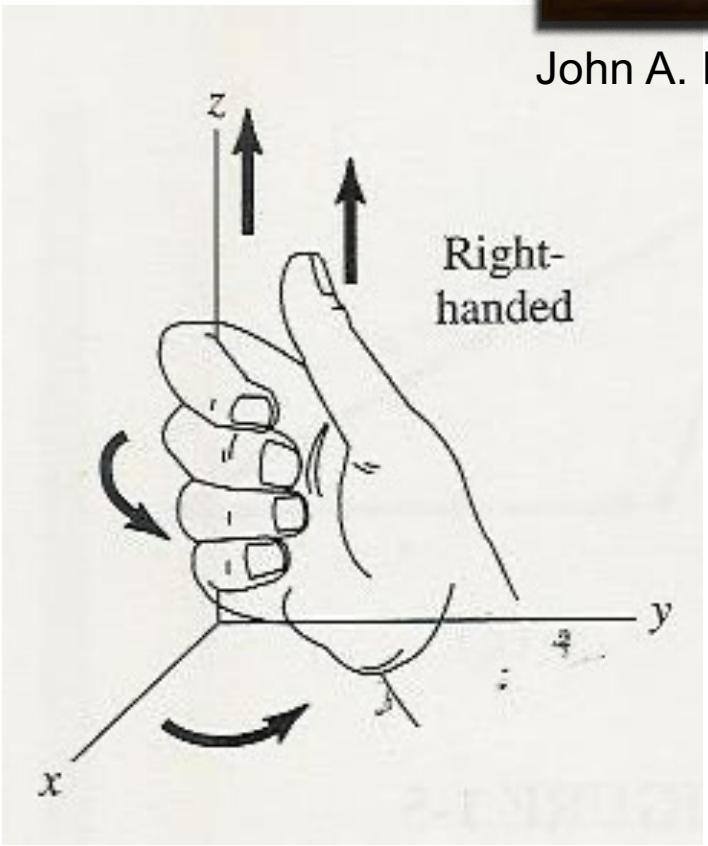
# (Review at home)

# Right Hand Rule



John A. Fleming



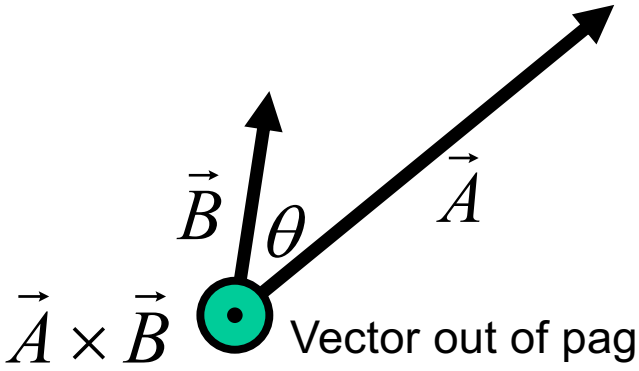
# (Review at home)
# Cross Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

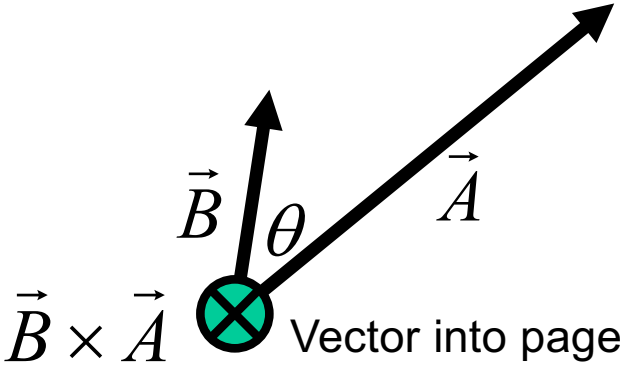
Cross Product of **A** and **B**

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

Same magnitude
Opposite direction




Vector out of page pointing at the classroom




Vector into page pointing **away** from classroom

# (Review at home)
# Cross Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

## Cross Product of **A** and **B**

$$\vec{A}\times\vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\sin\theta$$

$\vec{A}\perp\vec{B}$

$$\sin\theta = 1$$

$$\vec{A}\times\vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\hat{a}_N$$

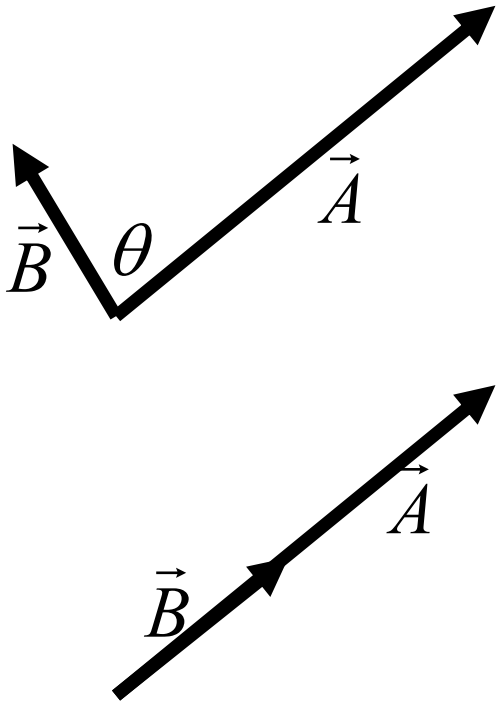
$$\vec{A}\perp\vec{B} \Leftrightarrow \vec{A}\times\vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\hat{a}_N$$

$\vec{A}\parallel\vec{B}$

$$\sin\theta = 0$$

$$\vec{A}\times\vec{B} = 0$$

$$\vec{A}\parallel\vec{B} \Leftrightarrow \vec{A}\times\vec{B} = 0$$



# (Review at home)
# Cross Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A}\times\vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\hat{a}_N$$

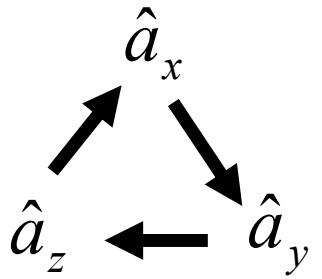
$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A}\times\vec{B} = 0$$

Cross Product of **Unit Vectors**

$$\hat{a}_x\times\hat{a}_x = 0 \qquad \hat{a}_y\times\hat{a}_x = -\hat{a}_z \qquad \hat{a}_z\times\hat{a}_x = \hat{a}_y$$

$$\hat{a}_x\times\hat{a}_y = \hat{a}_z \qquad \hat{a}_y\times\hat{a}_y = 0 \qquad \hat{a}_z\times\hat{a}_y = -\hat{a}_x$$

$$\hat{a}_x\times\hat{a}_z = -\hat{a}_y \qquad \hat{a}_y\times\hat{a}_z = \hat{a}_x \qquad \hat{a}_z\times\hat{a}_z = 0$$

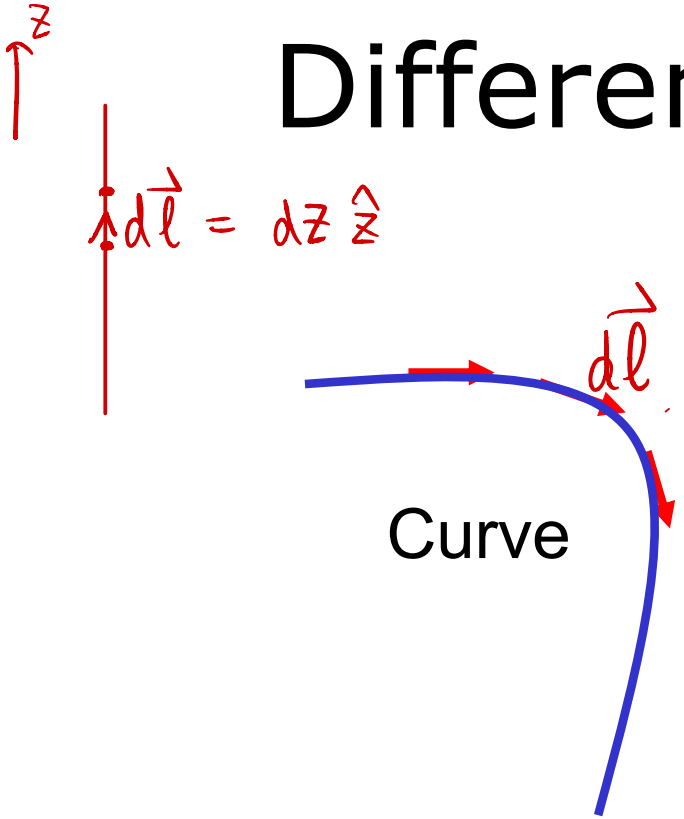



RH Coordinate System
Sign Convention

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_yB_z - A_zB_y)\hat{a}_x + (A_zB_x - A_xB_z)\hat{a}_y + (A_xB_y - A_yB_x)\hat{a}_z$$

Vector
Perpendicular to **A** & **B**
Right Hand Rule to Select Direction

# Differential Length Vector




Always tangent to a curve or surface

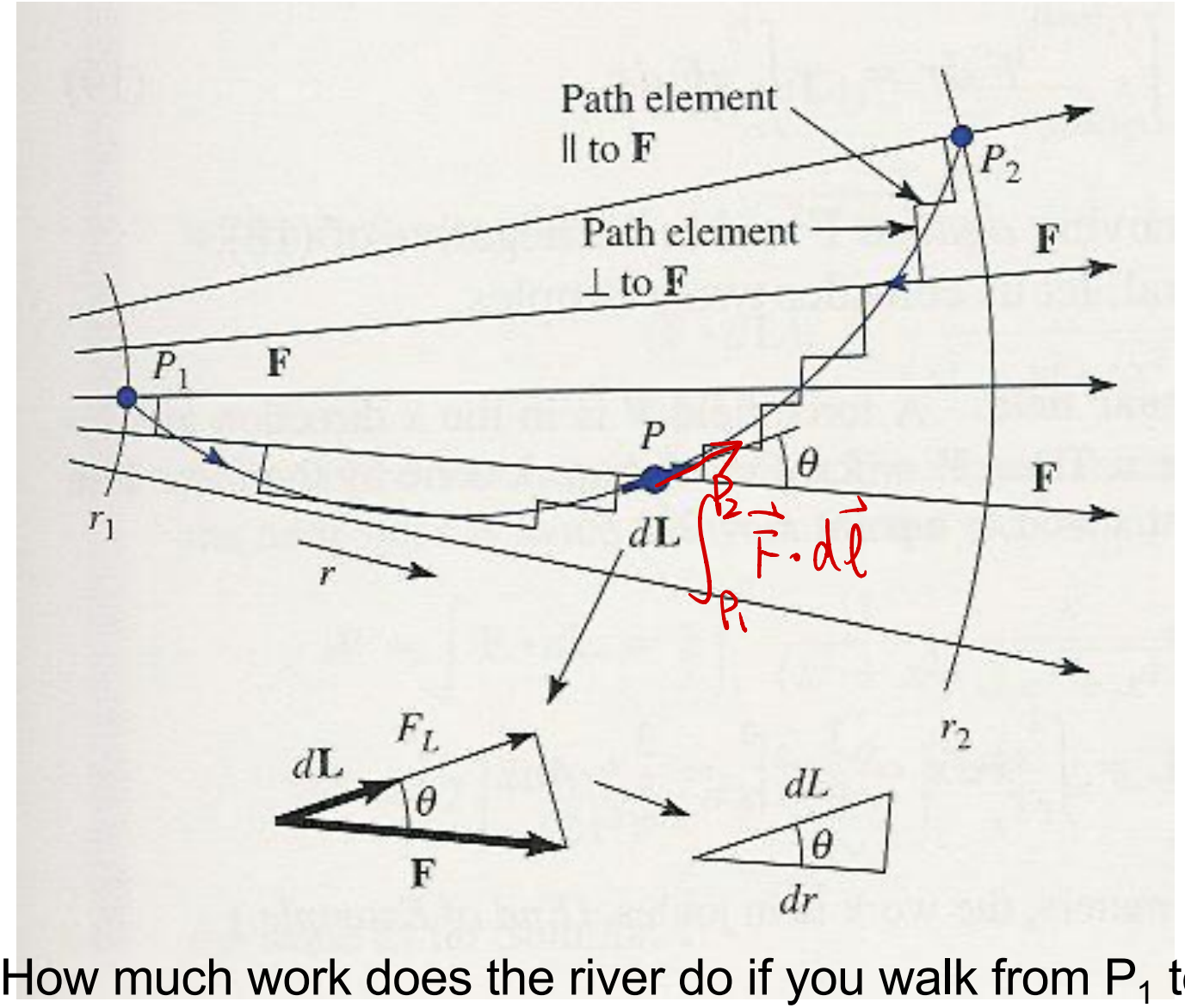
Exact vector is different at different places on the curve or surface

What we will plug into Faraday's law

$$d\vec{l} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$$

(in cartesian coordinates only)

Imagine you are *walking* in a river and the water pushes with a force, **F**, determined by the river's VELOCITY *VECTOR FIELD*.




How much work does the river do if you walk from $P_1$ to $P_2$ along the path shown?

# Discussion Problem

- Given **A**=$3\hat{x}+2\hat{y}+\hat{z}$, **B**= $\hat{x}+\hat{y}-\hat{z}$, **C**= $\hat{x}+2\hat{y}+3\hat{z}$, find:
  a. **B** x **C**
  b. **A** · (**B** x **C**)

a). $\vec{B}\times\vec{C} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & 1 & -1 \\ 1 & 2 & 3 \end{vmatrix} = \hat{x}(3+2) - \hat{y}(3+1) + \hat{z}(2-1) = 5\hat{x} - 4\hat{y} + \hat{z}$

b) $\vec{A}\cdot(\vec{B}\times\vec{C}) = \langle 3, 2, 1\rangle \cdot \langle 5, -4, 1\rangle = 15 - 8 + 1 = 8$

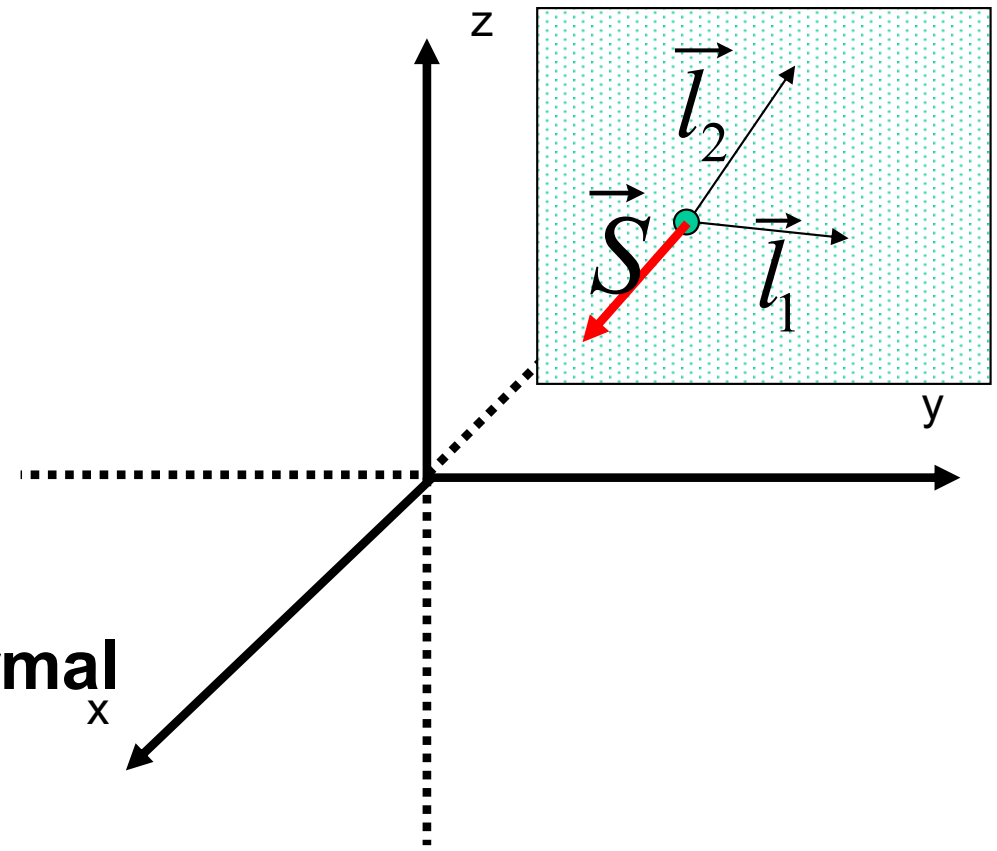
# Surface Vector

Easy example surface:
Flat surface in the yz plane

Pick a point on the surface

Find two vectors at that point that are tangent to the surface

A **surface vector** is always **normal** (perpendicular) to the surface




You get **normal** vectors by performing a **cross product**

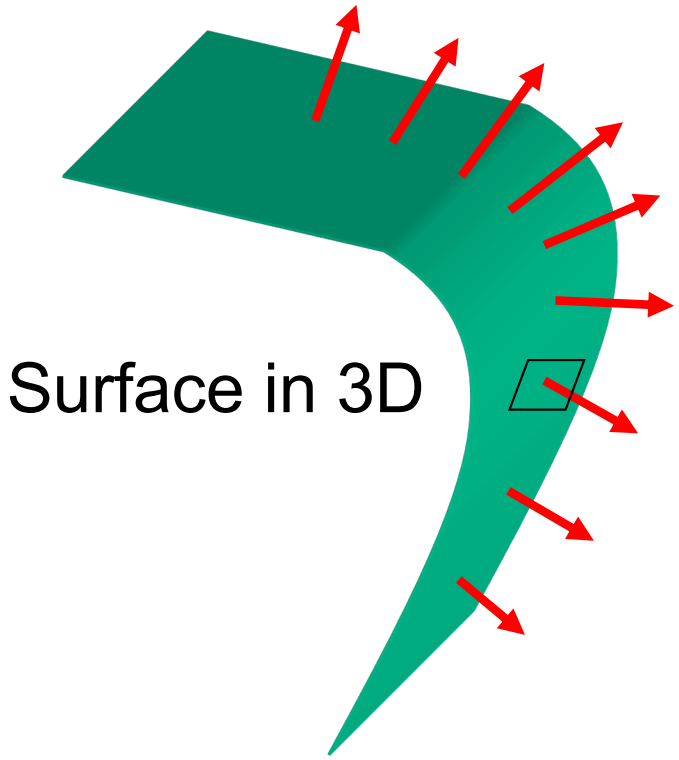
$$\vec{S} = \vec{l}_1 \times \vec{l}_2$$

out

2 1

into

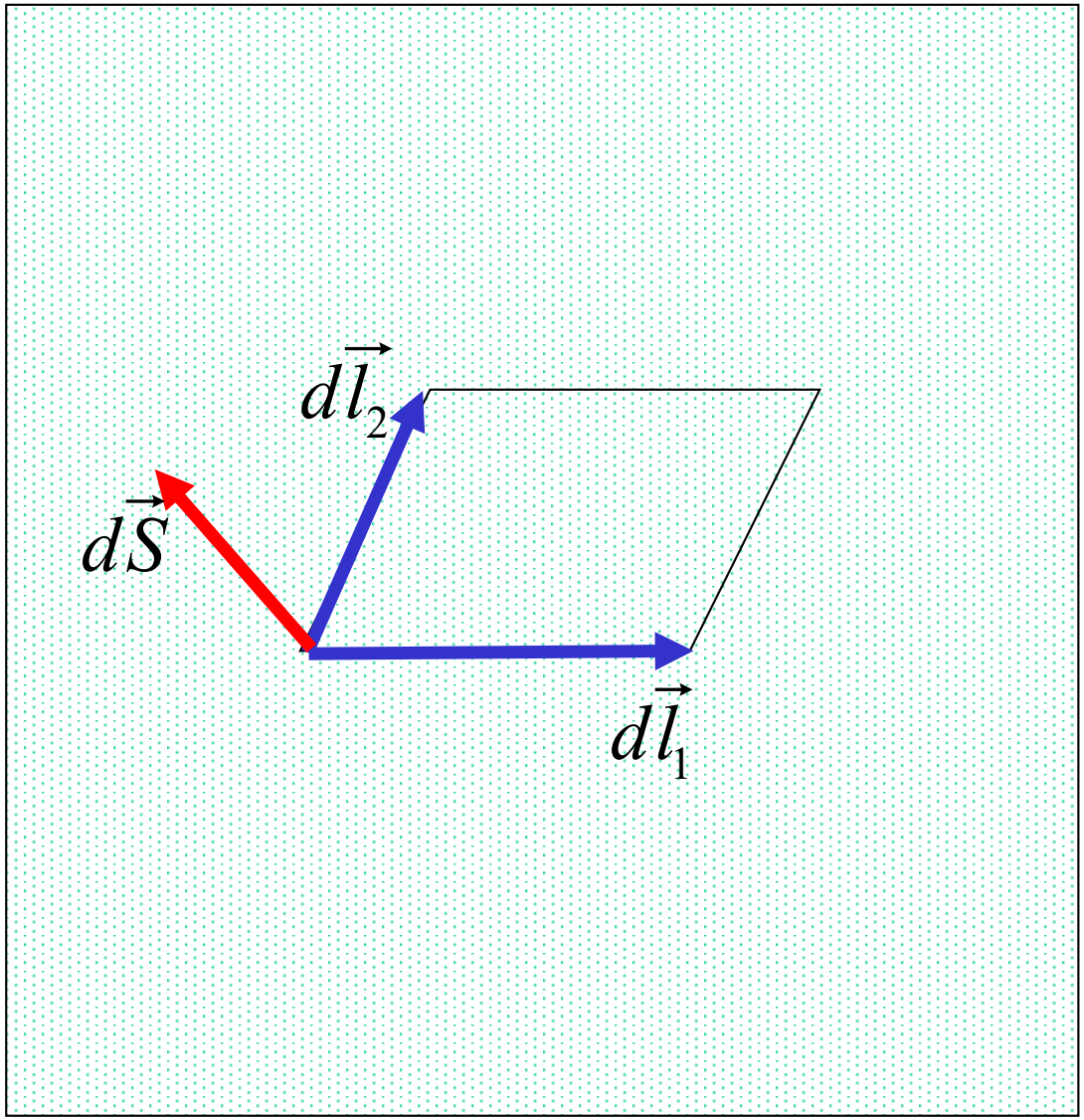
# Surface Vectors




How do we come up with a surface vector when a surface is curved?

If we zoom into a small enough area, the surface will “look” flat

# Differential Surface Vectors




Find two differential length vectors at the point on the surface.

Think of d**S** as a tiny parallelogram with sides bounded by d$\mathbf{l}_1$ and d$\mathbf{l}_2$ and direction perp to the surface.

$$d\vec{S} = d\vec{l}_1 \times d\vec{l}_2$$

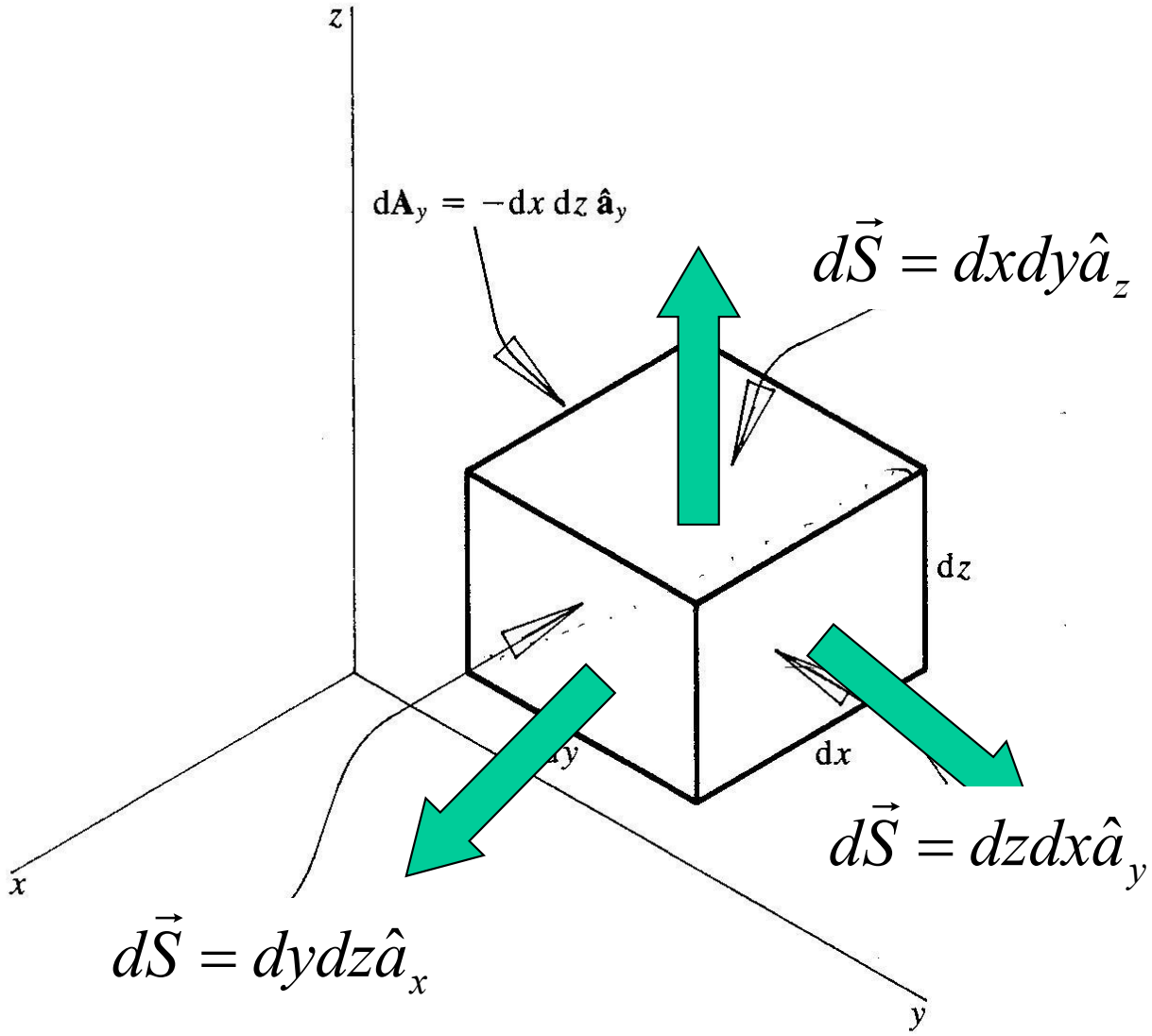
# Cartesian Coordinates

(x,y,z)

$$d\vec{l} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$$

$$d\vec{S} = \pm dydz\hat{a}_x$$
$$d\vec{S} = \pm dzdx\hat{a}_y$$
$$d\vec{S} = \pm dxdy\hat{a}_z$$

# Differential area or surface vectors for unit vectors in cartesian coords



# Lecture 1 Summary

- Dot Product $\vec{A} \bullet \vec{B} =$ _____ = _____
- Cross Product

$$\vec{A} \times \vec{B} = \begin{vmatrix} _ & _ & _ \\ _ & _ & _ \\ _ & _ & _ \end{vmatrix} = <_____, _____, _____>$$

- Differential vector

$$d\vec{l} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$$

$$d\vec{S} = \pm dydz\hat{a}_x$$
$$d\vec{S} = \pm dzdx\hat{a}_y$$
$$d\vec{S} = \pm dxdy\hat{a}_z$$

# Next

- Next two lectures:
  - Scalar and Vector Fields
  - The Lorentz Force
  - Coulomb's Law
  - Surf. Integrals/Gauss' Law